

HIT 137

Group Assignment 1 (10% Mark)

Please submit one .py file and ensure the file contains the following information:

Group Name: [Your Group Name]

Group Members:

[Member 1 Name] - [Member 1 ID]

[Member 2 Name] - [Member 2 ID]

[Member 3 Name] - [Member 3 ID]

[Member 4 Name] - [Member 4 ID]

This is a Group Assignment, each group only needs to submit once on the LearnLine.

Please ZIP your .py file and submit the ZIP file.

Late submissions: A late submission penalty of 5% of the total available marks per day will apply.

Introduction

You will build a text analyser. Take a block of text from user input, your program will read through it and report a range of statistics about it: how many letters and digits it contains, how many words and sentences, the longest word, the average word length, and finally a simple readability score that estimates how hard the text is to read.

Caveats

You may use standard Python 3 features covered in class so far.

You may not use the following shortcuts:

`str.count()`; `str.split()`; any third-party library.

You may freely use helpers such as:

`len()`, `.lower()`, `.upper()`, `.isalpha()`, `.isdigit()`, `.isspace()`, membership tests like `char in "aeiou"`, and the `+=` operator to build up a string.

Task 1: Character census

Scan the input text one character at a time and count:

1. the total number of characters,
2. the number of letters (a–z, A–Z),
3. the number of digits (0–9),
4. the number of whitespace characters (spaces, tabs, newlines),
5. the number of other characters (everything else — punctuation and so on).

Print a short, clearly labelled report of these five numbers. As a self-check, the four category counts should add up to the total.

Task 2: Case and vowel breakdown

Extend your program so that you also count:

1. uppercase letters and lowercase letters separately,
2. vowels and consonants separately (treat a e i o u as vowels, in either case),
3. how many times each individual vowel appears.

Add these to your report.

Task 3: Word statistics

Treat a word as a run of one or more letters (you may also allow an apostrophe inside a word, so that *we're* and *isn't* count as single words). Anything that is not part of a word (spaces, punctuation, digits, newlines) separates one word from the next.

Using this rule, report:

1. the total number of words,
2. the longest word and its length (if several words tie for longest, report the first one),
3. the average word length, rounded to one decimal place.

Task 4: Line and sentence analysis

Add three more measurements:

1. the number of lines in the text (a line is separated by the newline character `'\n'`),
2. the number of sentences, where a sentence ends with: `.`, `!`, or `?`,
3. the length of the longest line (in characters).